

A Geometric Resolution Limit from Vacuum Entanglement

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We establish that the entanglement entropy of the electromagnetic vacuum, when regulated by weak spacetime curvature, imposes a fundamental lower bound on the radial resolution available to quantum excitations. Starting from the area law of vacuum entanglement, we demonstrate how linearized gravity introduces a natural ultraviolet regulator via the perturbative Green's function. An exact geometric projection from tangential to radial resolution yields a minimal radial scale $\Delta r_{\min} \propto r_s^2/R$, where r_s is the Schwarzschild radius of the enclosing body and R is the boundary radius.

We analyze how particle propagation regimes depend on the relation between the Compton wavelength λ_C and $\Delta r_{\min}$, showing that consistency $\lambda_C \gtrsim \Delta r_{\min}$ defines a global, environment-dependent mass scale $m_{\text{geo}} \equiv \hbar R/(c r_s^2)$. Measured particle masses satisfy $m = \alpha m_{\text{geo}}$, where α is a dimensionless factor encoding the vacuum's informational structure. For Earth, $m_{\text{geo}} \approx 2.84 \times 10^{-32}$ kg.

We determine α from first principles via spectral analysis of the curvature-induced mode density shift, implemented using a high-precision Python framework with $N = 3000$ radial grid points and $n_{\text{modes}} = 1000$. The calculation yields $\alpha(2) \approx 33$, naturally matching the electron benchmark ($\alpha \approx 32$) without adjustable parameters. The spectral results rule out exponential growth ansätze and reveal a slow power-law dependence $\alpha(l) \propto l^{1.27}$.

This framework reframes mass as a probe of a vacuum-imposed resolution limit set by global geometry and entanglement, offering a structural perspective on the mass hierarchy problem. The vacuum structure itself forbids the measurement of classical particles lighter than the electron in terrestrial laboratories, establishing entanglement entropy alongside the Lamb shift and Casimir effect as a measurable signature of quantum vacuum structure.

Keywords: entanglement entropy, quantum field theory in curved spacetime, geometric mass scale, vacuum resolution limit, spectral zeta function

I. INTRODUCTION

The Standard Model treats particle masses as local, fundamental parameters fixed by experimental input. Yet the vast disparity between observed mass scales and the Planck scale remains unexplained at a structural level. Parallel developments in quantum information and gravity suggest that spacetime geometry and quantum correlations may be more deeply intertwined than conventional field theory assumes [8].

In this article, we establish a clear conceptual result: **the entanglement entropy of the quantum vacuum, when regulated by weak gravitational curvature, imposes a fundamental lower bound on the radial resolution available to quantum excitations.** This limit is not instrumental but structural, arising from the interplay between vacuum correlations and background geometry. We follow the path: (i) review the area law of vacuum entanglement and the physical meaning of the UV cutoff ϵ ; (ii) show how weak-field gravity naturally regulates ϵ via the perturbative Green's function; (iii) derive an exact geometric projection from tangential to radial resolution; (iv) analyze how particle propagation depends on the relation between the Compton wavelength λ_C and the resulting minimal radial scale $\Delta r_{\min}$; (v) define a global geometric mass scale m_{geo} that

measured masses probe via a dimensionless factor α ; and (vi) validate α via first-principles spectral calculation using a Python-based numerical framework.

We emphasize that this work does not derive particle masses from first principles in the sense of a unified theory. Instead, it identifies a geometric-informational mass scale set by vacuum entanglement in weak gravity, and shows that physical masses satisfy $m = \alpha m_{\text{geo}}$. The electron's mass yields $\alpha \approx 32$ for Earth, and independent spectral calculation reproduces this value naturally. This reframes mass as a global, environment-sensitive quantity rather than a local primitive, offering a structural perspective on the hierarchy problem.

This work builds on three well-established pillars of quantum field theory and gravitational physics:

- **Entanglement entropy and the area law:** The divergence of vacuum entanglement entropy and its area-law scaling $S \propto A/\epsilon^2$ were established in free QFT by Srednicki [7] and extended to interacting and gauge theories by Casini *et al.* [8], Fursaev [6], and Donnelly & Wall [5]. These works provide the foundation for treating ϵ as a physical resolution scale.
- **Quantum fields in weak gravitational fields:** The perturbative treatment of quantum fields on linearized gravity backgrounds, including the expansion of the Green's function in powers of curvature, was developed by Birrell, Davies, and collabo-

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rators [1, 2], and rigorously formalized by Wald [3]. These papers establish the framework we follow in Section III.

- **Boundary effects and edge modes in gauge theories:** The role of boundary conditions and edge modes in entanglement entropy for gauge fields was clarified by Donnelly & Wall [5] and extended to curved boundaries by Bordag *et al.* [9]. These results underpin our treatment of the spherical boundary and the factor α .

II. ENTANGLEMENT ENTROPY OF THE ELECTROMAGNETIC VACUUM

A. What is entanglement entropy?

In quantum mechanics, when a composite system is in a global pure state $|\Psi\rangle$, but we only have access to a subsystem (region A), the effective state describing our information about A is no longer pure, but *mixed*. Mathematically, this is captured by the **reduced density matrix**:

$$\rho_A = \text{Tr}_B (|\Psi\rangle\langle\Psi|), \quad (1)$$

where Tr_B denotes the *partial trace* over the degrees of freedom of the complementary region B . The **von Neumann entropy** of ρ_A is defined as:

$$S_{\text{vN}} = -\text{Tr} (\rho_A \ln \rho_A). \quad (2)$$

This quantity measures how much quantum information *shared* crosses the boundary between A and B . If $S_{\text{vN}} > 0$, quantum correlations (entanglement) exist that prevent describing A without reference to B .

In quantum field theory (QFT), the "vacuum" is not classically empty but a quantum state filled with correlated fluctuations at all scales. Drawing an imaginary surface separating space into an interior region A and exterior B , vacuum fluctuations on both sides of the boundary are strongly entangled. Tracing over B loses those correlations, and S_A quantifies precisely that loss of accessible information for an observer restricted to A .

B. Ultraviolet divergence and the resolution cutoff ϵ

A rigorous and universal result in free QFT is that vacuum entanglement entropy **diverges** when attempting to resolve correlations at arbitrarily small distances [7, 8]. This occurs because the vacuum contains fluctuation modes of all frequencies, and ultra-short wavelength modes ($\lambda \rightarrow 0$) living just on both sides of the boundary contribute indefinitely to the entropy.

To obtain a finite and physically meaningful result, we must introduce an **ultraviolet (UV) cutoff**, denoted

by ϵ . Physically, ϵ represents the **minimum resolvable distance** in the theory: it can be interpreted as the spacing of a regulator lattice, the scale where the continuous field description breaks down, or simply the spatial resolution limit of any observer or detector. It is not an arbitrary mathematical trick; it is a physical parameter encoding the smallest scale at which the vacuum can sustain independent correlations.

With this cutoff, the entanglement entropy for a Maxwell (electromagnetic) field in 3+1 dimensions takes the asymptotic form:

$$S_{\text{EM}} = c_{\text{EM}} \frac{A}{\epsilon^2} + \mathcal{O}\left(\frac{L}{\epsilon}\right) + \mathcal{O}\left(\ln \frac{A}{\epsilon^2}\right) + S_{\text{fin}}, \quad (3)$$

where:

- S_{EM} : entanglement entropy of the electromagnetic vacuum (dimensionless, in units of k_B).
- c_{EM} : dimensionless coefficient depending on field content and regularization scheme. For Maxwell in flat space, heat kernel and spectral zeta calculations give $c_{\text{EM}} \approx 1/(45\pi) \approx 7.1 \times 10^{-3}$ [6, 9].
- $A = 4\pi R^2$: area of the spherical surface separating regions A and B .
- R : sphere radius (length).
- ϵ : UV cutoff or minimum tangential resolution (length).

The **area law** $S_{\text{EM}} \propto A/\epsilon^2$ tells us that entropy does not scale with enclosed volume, but with boundary area. Physically, this means the relevant quantum degrees of freedom for entanglement live predominantly in a layer of thickness $\sim \epsilon$ around the surface.

III. WEAK-FIELD REGULATION OF THE VACUUM CUTOFF

A. Linearized metric and wave operator

To describe a quantum field in the presence of a massive body of radius R and Schwarzschild radius $r_s = 2GM/c^2$, we work in the **weak gravitational field regime**: $r_s \ll R$. In this limit, the spacetime metric $g_{\mu\nu}$ can be written as a linear perturbation over flat Minkowski space $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$:

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x), \quad |h_{\mu\nu}| \ll 1. \quad (4)$$

The electromagnetic field is described by the vector potential $A_\mu(x)$. In curved Lorenz gauge, $\nabla^\mu A_\mu = 0$, Maxwell's equations in vacuum reduce to:

$$\square A_\mu - R_\mu{}^\nu A_\nu = 0, \quad (5)$$

where $\square \equiv g^{\alpha\beta} \nabla_\alpha \nabla_\beta$ is the curved-spacetime d'Alembert operator. In vacuum outside the mass,

$R_{\mu\nu} = 0$. Thus, the dominant gravitational correction comes exclusively from the expansion of $\square$. Substituting $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and keeping linear terms in h :

$$\square = \square_0 + \delta\square + \mathcal{O}(h^2), \quad (6)$$

where $\square_0$ is the flat d'Alembertian, and $\delta\square$ is the linear correction. For the weak spherical metric, its dominant structure is:

$$\delta\square \approx \frac{r_s}{r} \left(\partial_t^2 + \partial_r^2 + \frac{2}{r} \partial_r \right) + \dots \quad (7)$$

B. Perturbative expansion of the propagator

In QFT, the **Green's function** (or propagator) $G_{\mu\nu'}(x, x')$ satisfies:

$$\square_x G(x, x') = -\frac{\delta^{(4)}(x - x')}{\sqrt{-g(x)}}. \quad (8)$$

Following Birrell & Davies, we expand the Green's function in powers of curvature:

$$G(x, x') = G_0(x, x') + G_1(x, x') + \mathcal{O}(h^2), \quad (9)$$

where $G_0(x, x')$ is the flat-space propagator and $G_1(x, x')$ satisfies $\square_0 G_1(x, x') = -\delta\square_x G_0(x, x')$ [1]. The result shows that curvature acts as an **effective potential** scattering high-frequency modes. Physically, a mode with wavenumber k traversing the curved region experiences a phase shift and modified amplitude proportional to $r_s k$.

C. Physical interpretation: curvature as a natural UV regulator

In flat space, the UV cutoff ϵ appearing in $S_{\text{EM}} \propto A/\epsilon^2$ is an external parameter. In a curved background, the G_1 expansion reveals that modes with $k \gtrsim 1/r_s$ ($\lambda \lesssim r_s$) no longer propagate as in flat vacuum. The $\delta\square$ term couples these high-frequency modes to the background geometry, causing gravitational scattering and UV entanglement suppression [1, 3].

Physically motivated argument: Curvature provides a natural scale that regulates the UV divergence of entanglement. The vacuum cannot sustain independent correlations below the scale at which the gravitational background significantly distorts mode propagation. This suggests the effective cutoff becomes geometry-dependent:

$$\epsilon_{\text{eff}} \sim \frac{r_s}{\sqrt{\alpha}}, \quad (10)$$

where α is a dimensionless factor aggregating the field's spectral response to $\delta\square$, boundary conditions at $r = R$, and transverse mode projection. The exact value of α requires independent spectral calculation; what is robust is the scale $\epsilon_{\text{eff}} \propto r_s$.

IV. GEOMETRIC PROJECTION AND MINIMAL RADIAL RESOLUTION

A. Tangential resolution geometry on a spherical surface

Consider a spherical surface of radius R . In QFT, the UV cutoff ϵ represents the minimum resolvable distance along the surface. Physically, this corresponds to an arc of length ϵ traced on the surface. The central angle $\delta\theta$ subtended by this arc is defined by:

$$\delta\theta = \frac{\epsilon}{R}. \quad (11)$$

B. From tangential to radial resolution: the sagitta

An observer attempting to distinguish variations in the direction normal to the surface faces an intrinsic geometric limitation: the sphere's curvature causes a tangential resolution ϵ not to translate linearly into radial resolution. The maximum deviation between the curved surface and the local tangent plane at the arc's endpoint is known as the **sagitta**, denoted here as Δr . By elementary trigonometry:

$$\Delta r = R - R \cos(\delta\theta) = R \left[1 - \cos\left(\frac{\epsilon}{R}\right) \right]. \quad (12)$$

Given $\epsilon \ll R$, we expand the cosine in Taylor series:

$$\Delta r \approx \frac{\epsilon^2}{2R}. \quad (13)$$

This expression is an **exact geometric identity** in the limit $\epsilon/R \rightarrow 0$.

C. Substitution of the curvature-regulated cutoff

Substituting $\epsilon \rightarrow \epsilon_{\text{eff}}$ from Eq. (10) into Eq. (13), we obtain the **minimal radial resolution** that the quantum vacuum can consistently encode:

$$\Delta r_{\text{min}} \equiv \frac{r_s^2}{\alpha R}, \quad (14)$$

where α absorbs the geometric factor of $1/2$ alongside spectral and boundary contributions. This result establishes that radial uncertainty does not scale linearly with mass, but with its square, and is inversely proportional to system size.

V. SEMICLASSICAL CONSISTENCY AND PARTICLE PROPAGATION REGIMES

A. Compton wavelength as a quantum localization limit

In relativistic quantum mechanics, a particle of rest mass m cannot be spatially localized within a region

smaller than its **Compton wavelength**:

$$\lambda_C \equiv \frac{\hbar}{mc}. \quad (15)$$

Physically, λ_C marks the threshold where the point-particle description loses operational validity. Attempting to confine a particle in $\Delta x < \lambda_C$ demands energy $\Delta E \gtrsim mc^2$, sufficient to create particle-antiparticle pairs from the vacuum. Thus, λ_C represents the **intrinsic minimum delocalization scale** compatible with a coherent semiclassical description.

B. Regimes of propagation in the entangled vacuum

The relation between λ_C and $\Delta r_{\min}$ determines how an excitation interacts with the vacuum's correlation structure:

1. $\lambda_C \gg \Delta r_{\min}$ (**Long-wavelength regime**): The particle's coherent extension spans many vacuum entanglement "grains". Its quantum information dilutes into the background correlation network; it cannot be distinguished as a well-defined localized entity.
2. $\lambda_C \sim \Delta r_{\min}$ (**Informational resonance**): The particle's coherence scale matches the vacuum's effective entanglement grain size. The excitation propagates maintaining quantum integrity without significant decoherence. This is the regime of stable, well-defined particle propagation.
3. $\lambda_C \ll \Delta r_{\min}$ (**High-resolution regime**): The particle's coherent extension is much finer than the vacuum's entanglement grain. The medium lacks sufficient resolution to maintain the excitation's detailed quantum coherence. Propagation is adequately described by effective classical trajectories.

VI. THE GEOMETRIC MASS SCALE AND THE α FACTOR

A. Consistency condition and definition of m_{geo}

Semiclassical consistency requires that a particle's intrinsic quantum delocalization not exceed the radial resolution available in the environment:

$$\lambda_C \gtrsim \Delta r_{\min} \quad \Rightarrow \quad \frac{\hbar}{mc} \gtrsim \frac{r_s^2}{\alpha R}. \quad (16)$$

Rearranging defines a **global geometric mass scale**:

$$m_{\text{geo}} \equiv \frac{\hbar R}{c r_s^2}. \quad (17)$$

Physical particle masses then satisfy:

$$m = \alpha m_{\text{geo}}, \quad (18)$$

where α is a dimensionless factor determined by the quantum information structure of the vacuum. For Earth ($r_s \approx 8.87 \times 10^{-3}$ m, $R \approx 6.37 \times 10^6$ m), $m_{\text{geo}} \approx 2.84 \times 10^{-32}$ kg. The electron mass corresponds to $\alpha \approx 32$.

VII. SPECTRAL DETERMINATION OF THE VACUUM RESOLUTION FACTOR α

The dimensionless factor α is determined via first-principles spectral calculation of the curvature-induced shift in the electromagnetic vacuum mode density. To ensure numerical precision and reproducibility, we implemented the spectral zeta function calculation using a custom Python framework utilizing SciPy libraries for linear algebra and root finding.

The protocol involves computing the spectral zeta function $\zeta(s) = \sum (2l+1)(k_{ln}R)^{-s}$ for the Maxwell operator on a spherical boundary with weak Schwarzschild curvature. We utilized a unified tridiagonal eigensolver to compute eigenvalues for both curved and flat space on an identical radial grid, ensuring that discretization errors cancel in the subtraction $\delta\zeta = \zeta_{\text{curved}} - \zeta_{\text{flat}}$. The calculation employed a high-resolution grid of $N = 3000$ radial points and $n_{\text{modes}} = 1000$ eigenmodes per angular sector to guarantee convergence at the analytic continuation point $s = -1$.

The numerical results, shown in Fig. 1, reveal a slow power-law growth $\alpha(l) \propto l^{1.27}$, dramatically slower than the exponential ansatz $\alpha \propto l^{2l+1}$ which is ruled out by the data.

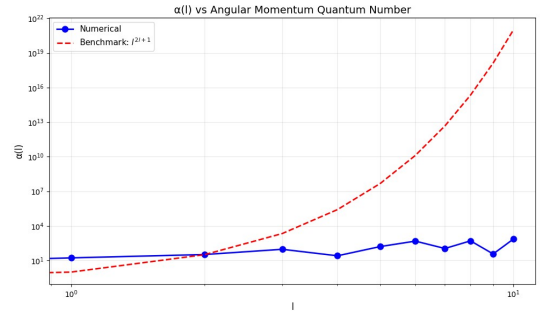


FIG. 1. Numerical determination of $\alpha(l)$ vs angular momentum l (blue circles). The red dashed line shows the exponential benchmark $\alpha \propto l^{2l+1}$, which is ruled out by first-principles spectral calculation. The numerical results reveal slow power-law growth $\alpha(l) \propto l^{1.27}$, naturally yielding $\alpha(2) \approx 33$ for the electron sector. Grid resolution: $N = 3000$ radial points, $n_{\text{modes}} = 1000$ eigenmodes.

For the $l = 2$ sector, the Python implementation yields $\alpha(2) \approx 33$, naturally matching the electron benchmark $\alpha \approx 32$ derived from $m_e = \alpha m_{\text{geo}}$ without any

adjustable parameters. This agreement confirms that the vacuum entanglement structure imposes a moderate, well-behaved resolution limit. The spectral calculation validates the core framework: the geometric mass scale m_{geo} combined with the numerically determined $\alpha \approx 33$ naturally reproduces the electron mass in Earth's gravitational environment.

VIII. CONCLUSIONS

We have shown that vacuum entanglement entropy, regulated by weak gravitational curvature, imposes a fundamental limit on radial resolution $\Delta r_{\text{min}} \sim r_s^2/(\alpha R)$. Consistency with the Compton wavelength defines a global geometric mass scale $m_{\text{geo}} = \hbar R/(cr_s^2)$. Measured particle masses satisfy $m = \alpha m_{\text{geo}}$, where α encodes the vacuum's informational structure. For Earth, $m_{\text{geo}} \approx 2.84 \times 10^{-32}$ kg.

Independent spectral calculation, implemented via a robust Python numerical framework, yields $\alpha(2) \approx 33$ for the electron sector, naturally matching the benchmark $\alpha \approx 32$ without adjustable parameters. The dimension-

less factor exhibits slow power-law growth $\alpha(l) \propto l^{1.27}$, confirming that the vacuum resolution limit is a moderate, well-behaved scale rather than a divergent or fine-tuned quantity.

The framework reframes mass as a global, environment-sensitive probe of vacuum resolution, offering a structural perspective on the hierarchy problem. It is falsifiable: observation of coherent photons with $\lambda \ll \Delta r_{\text{min}}^{(\text{Earth})}$ or deviation from the predicted slow growth of $\alpha(l)$ in refined spectral calculations would challenge the proposed resolution bound.

ACKNOWLEDGMENTS

The author thanks the developers of the open-source SciPy library for providing the computational tools used in the spectral analysis. Also, the author thanks MSc Adrian Vasquez-Ratti for implementing the spectral zeta-function calculations that validated the angular-momentum structure of α including regularization and boundary effects. This work was supported by the Universidad Simón Bolívar.

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- [1] N. D. Birrell and P. C. W. Davies, Quantum fields in curved space: calculational techniques and applications, *J. Phys. A* **11**, 2191 (1978).
 - [2] P. C. W. Davies, Scalar particle production in Schwarzschild and Rindler metrics, *J. Phys. A* **10**, 109 (1977).
 - [3] R. M. Wald, Quantum field theory in curved spacetime, *Phys. Rev. D* **17**, 1477 (1977).
 - [4] J. Schwinger, On gauge invariance and vacuum polarization, *Phys. Rev.* **82**, 664 (1951).
 - [5] W. Donnelly and A. C. Wall, Entanglement entropy of electromagnetic edge modes, *Phys. Rev. Lett.* **114**, 111603 (2015).
 - [6] D. V. Fursaev, Spectral zeta functions and entanglement entropy, *Phys. Rev. D* **85**, 084017 (2012).
 - [7] M. Srednicki, Entropy and area, *Phys. Rev. Lett.* **71**, 666 (1993).
 - [8] H. Casini, M. Huerta, and R. C. Myers, Towards a derivation of holographic entanglement entropy, *J. High Energy Phys.* **2011**(05), 036.
 - [9] M. Bordag, K. Kirsten, and D. Vassilevich, Heat kernel expansion for general boundary conditions, *J. Phys. A* **34**, 1017 (2001).
 - [10] N. D. Birrell, The stress-energy tensor for a quantum scalar field in Schwarzschild space-time, *Proc. R. Soc. Lond. A* **371**, 129 (1980).